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Effects of protein and amino acid supplementation on muscle protein metabolism in relation to exercise

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Abstract Exercise promotes protein and amino acid breakdown in skeletal muscle. Proper nutrient intake in relation to exercise is thus important to maintain or build up skeletal muscle. It has been demonstrated that supplementation of branched-chain amino acids (BCAAs) before exercise has beneficial effects on skeletal muscle, such as decreasing exercise-induced muscle damage and soreness. Protein synthesis in skeletal muscle is enhanced after exercise, and proper timing of supplementation of protein and amino acids for effectively stimulating muscle protein synthesis is herein discussed. Treatment of elderly individuals with exercise and nutrient supplementation is also demonstrated to have efficacy against age-related sarcopenia.

Keywords : resistance exercise, protein synthesis, supplement, essential amino acids, branched-chain amino acids

Introduction

Skeletal muscles with high power and flexibility are very important for participants in athletic competitions. Many athletes incorporate resistance training in their training programs to build up suitable skeletal muscle, and during such training, good nutrition in vitally important. As resistance training is also common in the general public for the purpose of health promotion, good nutrition is also important for public health.

In many developed countries, age-related sarcopenia characterized by a gradual, but progressive, loss in skeletal muscle mass and strength is a problem[1,2]. In the USA, medical expenses related to sarcopenia reached 18.5 billion dollars in 2000[3]. It is therefore important to maintain skeletal muscles in the elderly in order to avoid sarcopenia.

In this paper, the authors summarize the effects of exercise on protein and amino acid metabolism and discuss the most effective timing of nutrient intake in relation to exercise in order to increase the mass and strength of skeletal muscles.

Protein and amino acid metabolism in skeletal muscles

As muscle accounts for approximately 40% of body weight of humans and contains more or less 20% protein, muscle tissue acts as a reservoir of amino acids in the human body. Furthermore, muscle contains 3-5 g of free amino acids, which play an active role in protein metabolism[4,5]. Therefore, proteins and amino acids are very important nutrients to sustain or increase skeletal muscle mass in humans. However, both nutritional treatment and exercise training is necessary for development of skeletal muscles.

1. Increased protein breakdown by exercise

There are 3 major protease systems in muscles: 1) the lysosome system, 2) the ATP-dependent ubiquitin-proteasome system, and 3) the Ca^{2+}-dependent protease system[4]. Among these systems, the lysosome system is known to be activated by acute exercise. A very recent study using gene-manipulated mice demonstrated that autophagy, which is involved in the lysosome protease system, is important for exercise performance[6]. However, there is little information on the roles of protease systems in exercise.

It is known that intensive acute exercise causes muscle damage, resulting in increased serum creatine kinase release from skeletal muscles[7]. Therefore, enzyme activity in serum is commonly used as an index of muscle damage, which is caused by non-routine exercise, particularly in untrained individuals[8]. Muscle damage is accompanied by increased muscle protein breakdown. This protein breakdown is detected by monitoring blood urea concentrations. Urea is formed in the liver from ammonia derived from protein/amino acid degradation. It has been reported that the blood urea concentration increases proportionally with duration of exercise[9], thus suggesting that protein degradation is promoted by prolonged exercise.

On the other hand, leucine oxidation during exercise,

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which can be intensively examined using stable isotope-labeled leucine, was elevated in the early stages of exercise without an increase in urea concentration[10]. These findings suggest that protein breakdown during exercise is underestimated in analyses based on measurement of urea production, and that some amino acids (*e.g.,* leucine) are used as substrates for energy production, even in the early stages of acute exercise.

2. Increased amino acid breakdown by exercise

In the amino acid catabolic system, the carbon skeletons of the amino acids ultimately yield precursors and intermediates of the citric acid cycle (Fig. 1). Numerous types of amino acids, in excess, are catabolized in the liver, whereas some amino acids (Leu, Ile, Val, Ala, Asp and Glu) can be directly oxidized for energy production in muscles[4,5]. Branched-chain amino acids (BCAAs: Leu, Ile and Val) are typical amino acids that are catabolized in muscles, but not the liver, and Ala and Glu (particularly Glu/Gln) act as transporters of amino groups from the muscles to liver.

The BCAA catabolic pathway (Fig. 2) exclusively exists in mitochondria. The first 2 steps of the pathway are common to the 3 BCAAs, and the second step catalyzed by branched-chain α-keto acid dehydrogenase complex (BCKDC) is important for regulation of catabolism. BCKDC is regulated by a phosphorylation/dephosphorylation cycle; the E1α subunit of BCKDC is inactivated by phosphorylation and reactivated by dephosphorylation. The inactivation of BCKDC is catalyzed by branched-chain α-keto acid dehydrogenase kinase (BDK), and the activation by branched-chain α-keto acid dehydrogenase phosphatase (BDP).

BDK appears to be the main regulator of BCKDC activity, and it has been reported that almost all BCKDC in the skeletal muscles of rested rats exists in an inactive/phosphorylated state[11-13], suggesting that BCAA oxidation is very low in resting skeletal muscles. This was also confirmed in human skeletal muscles[14,15]. These findings suggest that the low BCKDC activity in resting skeletal muscle contributes to providing BCAAs for protein synthesis. It has been reported that chronic activation of BCKDC, by administration of a BDK inhibitor (clofibrate) into rats, induces a decrease in muscle proteins[16].

It has also been demonstrated that acute exercise significantly activates the muscle BCKDC in rats; after 2 hours of running on the treadmill (30 m/min), ~80% of muscle BCKDC was in an active state[11]. This enzyme activation is associated with increased BCAA oxidation in rats[12,17], and these phenomena were also confirmed in humans[18]. Hepatic BCKDC activity is low in rats fed a low-protein diet (BCAA-deficient condition), and this low activity was significantly elevated by acute exercise[19], thus suggesting that BCAA oxidation was promoted in the animals, even under BCAA-deficient conditions. Therefore, the BCAA requirement may be increased by exercise.

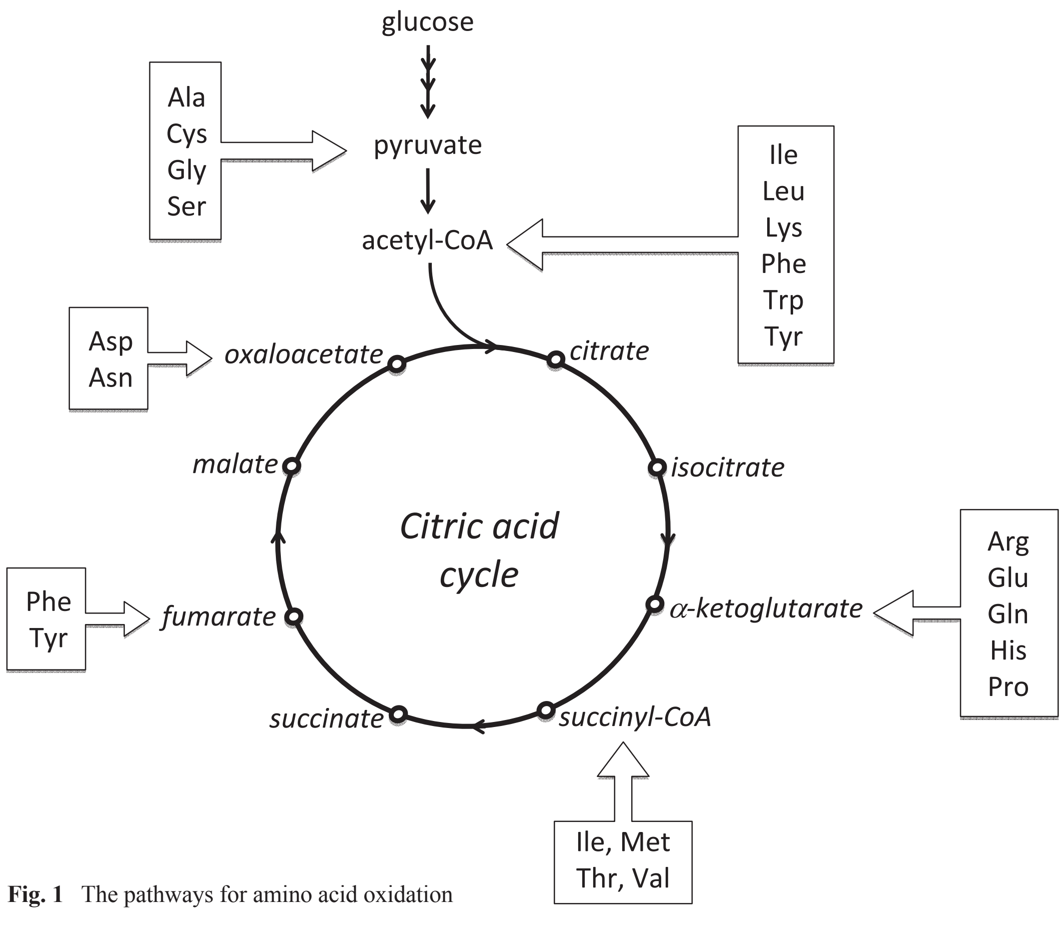

Fig. 1 The pathways for amino acid oxidation

Physiological effects of BCAA supplementation on skeletal muscles

1. Regulation of protein metabolism by leucine.

It has been demonstrated that leucine stimulates protein synthesis by promoting mRNA translation[20-22]. This mechanism involves activation (phosphorylation) of the mammalian target of rapamycin (mTOR, particularly mTOR complex 1), which phosphorylates ribosome protein S6 kinase 1 (S6K1), that subsequently phosphorylates ribosome protein S6 and eukaryotic initiation factor 4E-binding protein 1 (eIF4E-BP1). The latter allows eIF4E to form an eIF4F complex with eIF4A and eIF4G, and this complex is associated with mRNA in the translation processes. Furthermore, it has been reported that activated mTOR inhibits the autophagy system[23] and the ubiquitin-proteasome system[24]. From these findings, it is evident that leucine stimulates protein accretion in skeletal muscles.

2. Inhibition of exercise-induced muscle protein breakdown by BCAA supplementation.

MacLean et al.[25] reported evidence suggesting that skeletal muscle protein breakdown is suppressed by BCAA supplementation before exercise in humans, although muscle NH_3 production is increased during exercise. In this study, the subjects ingested a total of 77 mg BCAAs/kg BW (two doses of 38.5 mg/kg BW) 20-45 min before the onset of exercise comprising 70-75% of one-legged maximal knee extension for 60 min. These findings suggest that supplemented BCAAs are degraded in the muscle during exercise, resulting in suppression of BCAA mobilization from muscle proteins. This phenomenon is probably related to increased BCAA oxidation during exercise, and the relatively low concentrations of free BCAAs in the muscle (~0.65 mM) (4) and plasma (0.45 mM)[26].

3. Effects of BCAA supplementation on exercise-induced muscle damage and soreness.

Muscle damage is commonly assessed based on increases in serum creatine kinase activity, which is released from damaged muscle cells. BCAA supplementation has been reported to partially suppress the exercise-induced increase in blood creatine kinase activity[27,28].

Muscle damage induced by exercise is commonly associated with muscle soreness. Delayed-onset muscle soreness (DOMS), which is most strongly felt at 24 to 72 h after exercise, is caused by non-routine or strenuous exercise. The effects of BCAA supplementation on squat exercise-induced DOMS has been examined using untrained young women[29-31]; and these studies demonstrated that BCAA supplementation before exercise significantly reduces the level of DOMS. This effect has also been demonstrated in male subjects[32]. Furthermore, it has been reported that BCAA supplementation reduces exercise-induced muscle damage and improves exercise performance in the successive exercise program of athletes under fatigue conditions[33]. These findings support the notion that BCAA supplementation is useful for sports, although the

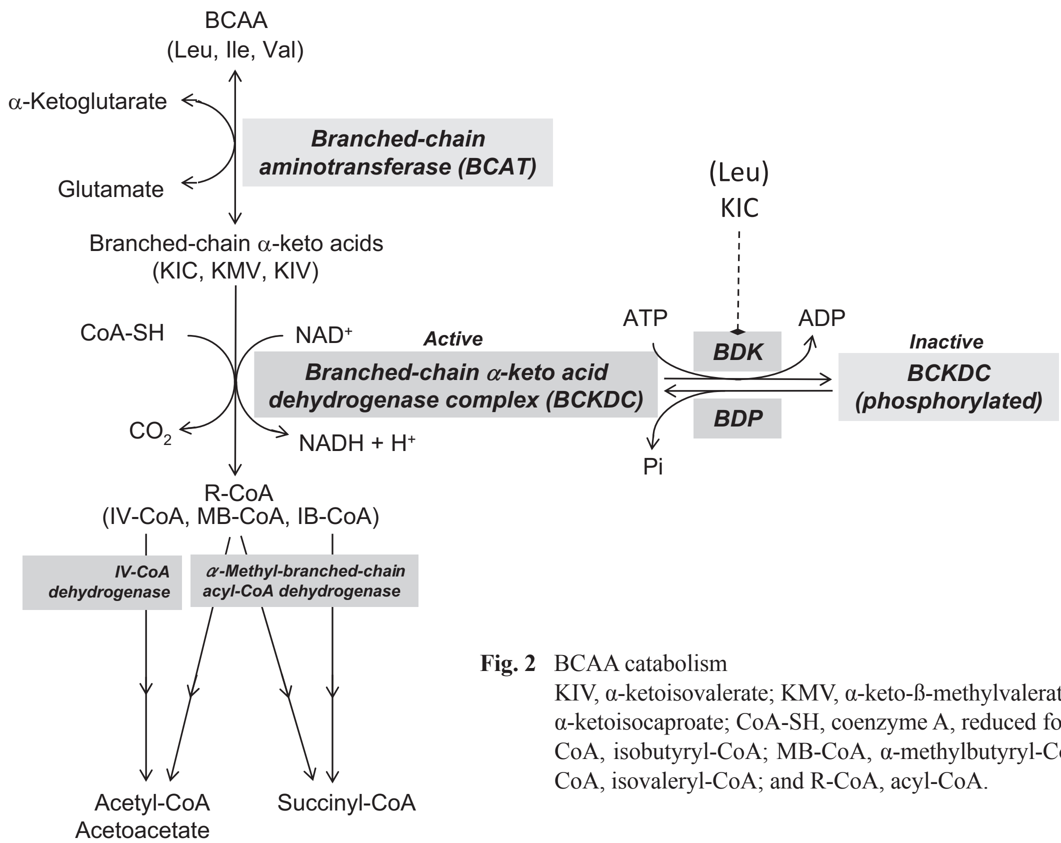

Fig. 2 BCAA catabolism
KIV, α-ketoisovalerate; KMV, α-keto-ß-methylvalerate; KIC, α-ketoisocaproate; CoA-SH, coenzyme A, reduced form; IB-CoA, isobutyryl-CoA; MB-CoA, α-methylbutyryl-CoA; IV-CoA, isovaleryl-CoA; and R-CoA, acyl-CoA.

detailed mechanism responsible for the action of BCAA remains to be clarified. However, it may be related to the regulatory functions of amino acids on muscle protein metabolism.

The other physiological functions of BCAA supplementation in relation to exercise are as follows. BCAA supplementation:
1) decreases central fatigue during exercise in humans[34,35],
2) spares muscle glycogen during exercise in rats[36], and
3) prolongs endurance exercise during heat stress in humans[37] and under normal conditions in rats[38].

An interesting study demonstrated a protective effect of BCAA supplementation against muscle cramps in patients with liver cirrhosis[39]. In Japan, a BCAA formulation is used in liver cirrhosis patients to improve decreases in plasma albumin levels. It is commonly observed that plasma BCAA levels are significantly lower in such patients, corresponding to decreased plasma albumin levels. Therefore, BCAA administration to patients is effective in improving albumin levels. Although the mechanisms responsible for the protective effects of BCAAs against muscle cramp are uncertain, it is thought that BCAA supplementation could suppress muscle protein breakdown by increasing the free BCAA concentrations in the tissue. This hypothesis may also be applied to muscle cramps during exercise; BCAA supplementation may be useful in suppressing muscle cramps during sporting events.

Protein and amino acid supplementation in relation to exercise to promote muscle protein synthesis

1. Protein requirements

It is known that exercise training and sufficient levels of good quality proteins are required for both athletes and untrained individuals to build strong muscles. Resistance training, such as weight training, is known to be particularly effective in building skeletal muscle. Recent studies have demonstrated that relatively mild intensity exercise training with blood flow restriction is an effective means to increase muscle mass by elevating expression of muscle proteins[40]. Good quality proteins have amino acid compositions that are necessary as nutrients for humans, and many animal proteins (typical examples: milk and egg proteins) are considered to be good proteins. It is generally recognized that protein requirements increase with intensive exercise, although there is no consensus. However, as intensive exercise promotes protein and amino acid breakdown during exercise and protein synthesis after exercise, it is recommended that athletes increase protein intake by 1.5 to 2-fold during intensive exercise training periods[4,8].

2. Timing of protein and amino acid supplementation

Recent research in the exercise nutrition field has examined the appropriate timing of protein or amino acid intake to promote muscle protein synthesis in relation to exercise training. These studies are herein summarized in an effort to determine the best timing of nutrient intake in relation to exercise.

As described above, protein and amino acid breakdown is promoted during exercise. It has been confirmed that muscle protein synthesis increases during the resting recovery period after exercise[4,41], and that this increased protein synthesis lasts for at least 2 days after exercise[41,42]. This suggests that nutrient intake after exercise should effectively increase protein synthesis. Experimenting with dogs[43], the muscle protein synthesis rate was found to be more effectively elevated by intravenous administration of amino acids and glucose at the end of (right after) exercise than at 2 h post-exercise. This finding was later confirmed in human studies.

Levenhagen et al.[44] measured the rates of muscle protein synthesis and breakdown during the post-exercise period in subjects aged 30-33 years, who were studied twice, with the same oral supplement (10 g of protein, 8 g of carbohydrate, and 3 g of fat) being administered either immediately after, or at 3 h after, 60 min of moderate-intensity exercise. The results clearly showed that the net protein synthesis rate was significantly higher in the former than in the latter cases. Esmarck et al.[45] demonstrated, using elderly subjects (around 74 years of age), that protein supplementation immediately after - as compared to that at 2-h after - resistance exercise was more effective in increasing muscle strength and muscle mass. Subjects participated in a 12-wk resistance training program (3 times per week) receiving oral protein in liquid form (10 g of protein, 7 g of carbohydrate, and 3 g of fat) immediately after, or at 2 h after, each training session. Based on these findings, protein supplementation immediately after exercise is recommended to build up skeletal muscle.

The effects of amino acid supplementation (6 g essential amino acid mixture and 35 g sugar) after resistance exercise on muscle protein synthesis has also been examined, and clearly shows that supplement ingestion at either 1 h or 3 h after resistance exercise equally increases muscle protein synthesis[46]. On the other hand, when this amino acid supplement was ingested immediately before and after resistance exercise, and comparisons made[47], results showed that increased protein synthesis was significantly higher in the former than in the latter cases, thus suggesting that amino acid supplementation immediately before resistance exercise is more effective at building skeletal muscle. The efficacy of protein supplementation before or after exercise was also examined using whey protein, and the results showed no differences in protein synthesis[48]. Therefore, the beneficial effects immediately before exercise are greater with amino acid supplementation than with protein supplementation. This may be due to digestion not being necessary for amino acids, resulting in more rapid absorption from the gut. In contrast to the beneficial effects of amino acid supplementation immediately before exercise, it has been reported that ingestion

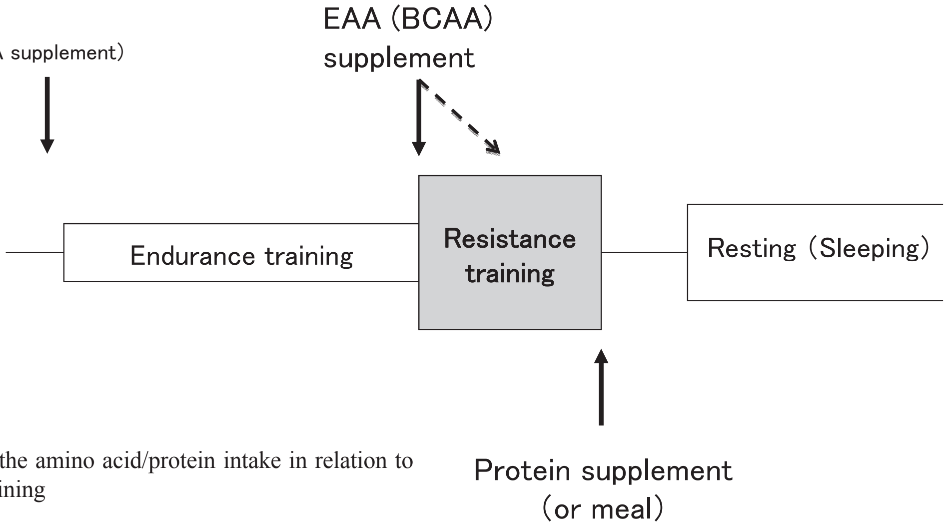

Fig. 3 Timing for the amino acid/protein intake in relation to exercise training

EAA, essential amino acid; and BCAA, branched-chain amino acid.
It is recommended to ingest the EAA (or BCAA) supplement right before and during resistance training and the protein supplement (or meal with enough protein) after the training in order to enhance muscle protein anabolism during the post-exercise resting period. The BCAA supplement may be effective to decrease central fatigue during the endurance training (34).

of essential amino acid supplement at 1 h before resistance exercise did not significantly increase muscle protein synthesis during the post-exercise recovery period[49]. The reason for this result is unclear, but it may be due to quick absorption of amino acids into circulation from the gut, and the short peak period for plasma concentrations.

In summary, the best time point for amino acid and protein supplementation, in relation to a training program, is shown in Fig. 3. Amino acid supplementation (particularly an essential amino acid mixture) should be ingested immediately before resistance exercise. When resistance exercise lasts more than 1 h, additional amino acid supplements should be ingested during training, because the peak of plasma amino acid concentration after ingestion of the supplement appears at approximately 30 min after ingestion and then decreases gradually. BCAAs in the essential amino acid mixture may be the main components responsible for physiological functions such as stimulation of muscle protein synthesis. After resistance training, protein supplements and/or a meal containing sufficient levels of proteins should be ingested without delay.

Treatment of elderly individuals with amino acid supplements and resistance training to ameliorate age-related sarcopenia

Aging induces insulin resistance, which promotes age-related sarcopenia[50]. As noted above, sarcopenia is a serious problem in many developed countries, but nutritional treatment with amino acid supplements and exercise training have been demonstrated to be effective in the treatment of sarcopenia[51,52]. Resistance exercise training with blood flow restriction is also effective for improving sarcopenia[53]. However, these studies have mainly been conducted in the USA, and evidence for such treatments in elderly Japanese is lacking.

References

1) Fujita S, Volpi E. 2006. Amino acids and muscle loss with aging. *J Nutr* 136: 277S-280S.
2) Meng SJ, Yu LJ. 2010. Oxidative stress, molecular inflammation and sarcopenia. *Int J Mol Sci* 11: 1509-1526.
3) Janssen I, Shepard DS, Katzmarzyk PT, Roubenoff R. 2004. The healthcare costs of sarcopenia in the United States. *J Am Geriatr Soc* 52: 80-85.
4) Rennie M. 1996. Influence of exercise on protein and amino acid metabolism. In: Rowell LB and Shepherd JT, eds, Handbook of Physiology, Section 12: Exercise: Regulation and Integration of Multiple Systems, Oxford University Press, New York, pp. 995-1035.
5) Wagenmakers AJM. 1998. Muscle amino acid metabolism at rest and during exercise: role in human physiology and metabolism. *Exerc Sport Sci Rev* 26: 287-314.
6) He C, Bassik MC, Moresi V, Sun K, Wei Y, Zou Z, An Z, Loh J, Fisher J, Sun Q, Korsmeyer S, Packer M, May HI, Hill JA, Virgin HW, Gilpin C, Xiao G, Bassel-Duby R, Scherer PE, Levine B. 2012. Exercise-induced BCL2-regulated autophagy is required for muscle glucose homeostasis. *Nature* 481: 511-515.
7) Armstrong RB, Ogilvie RW, Schwane JA. 1983. Eccentric exercise-induced injury to rat skeletal muscle. *J Appl Physiol* 54: 80-93.
8) Evans WJ, Meredith CN, Cannon JG, Dinarello CA, Frontera WR, Hughes VA, Jones BH, Knuttgen HG.1986. Metabolic changes following eccentric exercise in trained and untrained men. *J Appl Physiol* 61: 1864-1868.
9) Poortmans JR. 1988. Protein metabolism. In: Poortmans JR, ed., Principles of Exercise Biochemistry, Medicine and Sport Science, Vol. 27, Karger, Basel, pp. 164-193.
10) Wolfe RR, Goodenough RD, Wolfe MH, Royle GT, Nadel ER.1982. Isotopic analysis of leucine and urea metabolism in exercising humans. *J Appl Physiol* 52: 458-466.
11) Shimomura Y, Suzuki T, Saitoh S, Tasaki Y, Harris RA, Suzuki M. 1990. Activation of branched-chain α-keto acid dehydrogenase complex by exercise: effect of high-fat diet

intake. *J Appl Physiol* 68: 161-165.

12) Shimomura Y. 1996. Regulation of branched- chain α-keto acid dehydrogenase complex in rat liver and skeletal muscle by exercise and nutrition. In: Patel MS, Roche TE, Harris RA, eds., Alpha-Keto Acid Dehydro-genase Complexes, Birkhauser Verlag, Basel, pp. 177-186.
13) Shimomura Y, Fujii H, Suzuki M, Fujitsuka N, Naoi M, Sugiyama S, Harris RA.1993. Branched-chain 2-oxo acid dehydrogenase complex activation by tetanic contractions in rat skeletal muscle. *Biochim Biophys Acta* 1157: 290-296.
14) Wagenmakers AJ, Brookes JH, Coakley JH, Reilly T, Edwards RH. 1989. Exercise-induced activation of the branched-chain 2-oxo acid dehydrogenase in human muscle. *Eur J Appl Physiol* 59: 159-167.
15) Suryawan A, Hawes JW, Harris RA, Shimomura Y, Jenkins AE, Hutson SM. 1998. A molecular model of human branched-chain amino acid metabolism. *Am J Clin Nutr* 68: 72-81.
16) Paul H, Adibi SA. 1979. Paradoxical effects of clofibrate on liver and muscle metabolism in rats. *J Clin Invest* 64: 405-412.
17) Hood DA, Terjung RL. 1987. Leucine metabolism in perfused rat skeletal muscle during contractions. *Am J Physiol* 253: E636-E647.
18) van Hall G, MacLean DA, Saltin B, Wagenmakers AJ. 1996. Mechanisms of activation of muscle branched-chain α-keto acid dehydrogenase during exercise in man. *J Physiol* 494: 899-905.
19) Kobayashi R, Shimomura Y, Murakami T, Nakai N, Otsuka M, Arakawa N, Shimizu K, Harris RA. 1999. Hepatic branched-chain α-keto acid dehydrogenase complex in female rats; activation by exercise and starvation. *J Nutr Sci Vitaminol* 45: 303-309.
20) Proud CG. 2007. Signalling to translation: how signal transduction pathways control the protein synthetic machinery. *Biochem J* 403: 217-234.
21) Kimball SR. 2007. The role of nutrition in stimulating muscle protein accretion at the molecular level. *Biochem Soc Trans* 35: 1298-1301.
22) Proud CG. 2009. mTORC1 signalling and mRNA translation. *Biochem Soc Trans* 37: 227-231.
23) Kadowaki M, Karim MR, Carpi A, Miotto G. 2006. Nutrient control of macroautophagy in mammalian cells. *Mol Aspects Med* 27: 426-443.
24) Herningtyas EH, Okimura Y, Handayaningsih AE, Yamamoto D, Maki T, Iida K, Takahashi Y, Kaji H, Chihara K.2008. Branched-chain amino acids and arginine suppress MaFbx/atrogin-1 mRNA expression via mTOR pathway in C2C12 cell line. *Biochim Biophys Acta* 1780: 1115-1120.
25) MacLean DA, Graham TE, Saltin B, 1994. Branched-chain amino acids augment ammonia metabolism while attenuating protein breakdown during exercise. *Am J Physiol* 267: E1010-E1022.
26) Ahlborg G, Felig P, Hagenfeldt L, Hendler R, Wahren J. 1974. Substrate turnover during prolonged exercise in man. *J Clin Invest* 53: 1080-1090.
27) Coombes JS, McNaughton LR. 2000. Effects of branched-chain amino acid supplementation on serum creatine kinase and lactate dehydrogenase after prolonged exercise. *J Sports Med Phys Fitness* 40: 240-246.
28) Nosaka K. 2003. Muscle soreness and amino acids. *Training Journal* 289: 24-28 (in Japanese).
29) Shimomura Y, Yamamoto Y, Bajotto G, Sato J, Murakami T, Shimomura N, Kobayashi H, Mawatari K. 2006. Nutraceutical effects of branched-chain amino acids on skeletal muscle. *J Nutr* 136: 529S-532S.
30) Sato J, Yamamoto Y, Hamada K, Shimomura Y. 2005. Effects of branched-amino acid drink on exercise-induced muscle soreness and fatigue. *J Clin Sports Med* 22: 837-839 (in Japanese).
31) Shimomura Y, Inaguma A, Watanabe S, Yamamoto Y, Muramatsu Y, Bajotto G, Sato J, Shimomura N, Kobayashi H, Mawatari K.2010. Branched-chain amino acid supplementation before squat exercise and delayed-onset muscle soreness. *Int J Sport Nutr Exerc Metab* 20: 236-244.
32) Jackman SR, Witard OC, Jeukendrup AE, Tipton KD. 2010. Branched-chain amino acid ingestion can ameliorate soreness from eccentric exercise. *Med Sci Sports Exerc* 2: 962-970.
33) Skillen RA, Testa M, Applegate EA, Heiden EA, Fascetti AJ, Casazza GA. 2008. Effects of an amino acid-carbohydrate drink on exercise performance after consecutive-day exercise bouts. *Int J Sport Nutr Exerc Metab* 18: 473-492.
34) Blomstrand E, Hassmén P, Ek S, Ekblom B, Newsholme EA. 1997. Influence of ingesting a solution of branched-chain amino acids on perceived exertion during exercise. *Acta Physiol Scand* 159: 41-49.
35) Wurtman RJ. 1983. Behavioural effects of nutrients. *Lancet,* 1145-1147.
36) Shimomura Y, Murakami T, Nakai N, Nagasaki M, Obayashi M, Li Z, Xu M, Sato Y, Kato T, Shimomura N, Fujitsuka N, Tanaka K, Sato M. 2000. Suppression of glycogen consumption during acute exercise by dietary branched-chain amino acids in rats. *J Nutr Sci Vitaminol* 46: 71-77.
37) Mittleman KD, Ricci MR, Bailey SP. 1998. Branched-chain amino acids prolong exercise during heat stress in men and women. *Med Sci Sports Exerc* 30: 83-91.
38) Calders P, Pannier JL, Matthys DM, Lacroix EM. 1997. Pre-exercise branched-chain amino acid administration increases endurance performance in rats. *Med Sci Sports Exerc* 29: 1182-1186.
39) Sako K, Imamura Y, Nishimata H, Tahara K, Kubozono O, Tsubouchi H. 2003. Branched-chain amino acids supplements in the late evening decrease the frequency of muscle cramps with advanced hepatic cirrhosis. *Hepatol Res* 26: 327-329, 2003.
40) Drummond MJ, Fujita S, Abe T, Dreyer HC, Volpi E, Rasmussen BB. 2008. Human muscle gene expression following resistance exercise and blood flow restriction. *Med Sci Sports Exerc* 40: 691-698.
41) Goodman MM. 1988. Amino acid and protein metabolism. In: Horton ES, Terjung RL, eds, Exercise, Nutrition, and Energy Metabolism, Macmillan, New York, pp. 89-99.
42) Walker DK, Dickinson JM, Timmerman KL, Drummond MJ, Reidy PT, Fry CS, Gundermann DM, Rasmussen BB. 2011. Exercise, amino acids, and aging in the control of human muscle protein synthesis. *Med Sci Sports Exerc* 43: 2249-2258.
43) Okamura K, Doi T, Hamada K, Sakurai M, Matsumoto K, Imaizumi K, Yoshioka Y, Shimizu S, Suzuki M. 1997. Effect of amino acid and glucose administration during postexercise recovery on protein kinetics in dogs. *Am J Physiol* 272:

E1023-E1030.
44) Levenhagen DK, Gresham JD, Carlson MG, Maron DJ, Borel MJ, Flakoll PJ. 2001. Postexercise nutrient intake timing in humans is critical to recovery of leg glucose and protein homeostasis. *Am J Physiol* 280: E982-E993.
45) Esmarck B, Andersen JL, Olsen S, Richter EA, Mizuno M, Kjaer M. 2001. Timing of postexercise protein intake is important for muscle hypertrophy with resistance training in elderly humans. *J Physiol* 535: 301-311.
46) Rasmussen BB, Tipton KD, Miller SL, Wolf SE, Wolfe RR. 2000. An oral essential amino acid-carbohydrate supplement enhances muscle protein anabolism after resistance exercise. *J Appl Physiol* 88: 386-392.
47) Tipton KD, Rasmussen BB, Miller SL, Wolf SE, Owens-Stovall SK, Petrini BE, Wolfe RR. 2001. Timing of amino acid-carbohydrate ingestion alters anabolic response of muscle to resistance exercise. *Am J Physiol* 281: E197-E206.
48) Tipton KD, Elliott TA, Cree MG, Aarsland AA, Sanford AP, Wolfe RR. 2007. Stimulation of net protein synthesis by whey protein ingestion before and after exercise. *Am J Physiol Endocrynol Metab,* 292: E71-E76.
49) Fujita S, Dreyer HC, Drummond MJ, Glynn EL, Volpi E, Rasmussen BB. 2009. Essential amino acid and carbohydrate ingestion before resistance exercise does not enhance postexercise muscle protein synthesis. *J Appl Physiol* 106: 1730-1739.
50) Fujita S, Glynn EL, Timmerman KL, Rasmussen BB, Volpi E. 2009. Superphysiological hyperinsulinaemia is necessary to stimulate skeletal muscle protein anabolism in older adults; evidence of a true age-related insulin resistance of muscle protein metabolism. *Diabetologia* 52: 1889-1898.
51) Katsanos CS, Kobayashi H, Sheffield-Moore M, Aarsland A, Wolfe RR. 2006. A high proportion of leucine is required for optimal stimulation of the rate of muscle protein synthesis by essential amino acids in the elderly. *Am J. Physiol Endocrynol Metabol* 291: E381-E387.
52) Fujita S, Rasmussen BB, Cadenas JG, Drummond MJ, Glynn EL, Sattler FR, Volpi E. 2007. Aerobic exercise overcomes the age-related insulin resistance of muscle protein metabolism by improving endothelial function and Akt/mammalian target of rapamycin signaling. *Diabetes* 56: 1615-1622.
53) Fry CS, Glynn EL, Drummond MJ, Timmerman KL, Fujita S, Abe T, Dhanani S, Volpi E, Rasmussen BB. 2010. Blood flow restriction exercise stimulates mTORC1 signaling and muscle protein synthesis in older men. *J Appl Physiol* 108: 1199-1209.